Supplementary Information

A catalog of archaeal HTH and RHH DNA-binding-domain families outside a nineteen-family reference set

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Supplementary Methods

Operational scope and terminology

The historical discovery universe comprised archaeal proteins detected by profiles in Pfam clans CL0123 and CL0057. Outside the reference set means not explained by the nineteen-family assignment used in that analysis. The exact mapping of families to twenty reference models, the 747 candidate-model accessions and the Pfam release were not preserved. The term therefore does not mean previously undescribed in all databases and does not establish transcription-factor activity.

Panel construction

Assemblies were assigned to seven mutually exclusive historical categories: Euryarchaeota, Crenarchaeota within TACK, Thaumarchaeota within TACK, other deep TACK, Korarchaeota within TACK, DPANN and Asgardarchaeota. All genomes from five underrepresented groups were retained. Euryarchaeota and Crenarchaeota within TACK were capped at 140 and 70. The exact GTDB release was not preserved.

Reciprocal exclusion

For each protein, the best-scoring candidate-domain interval was compared with the best canonical hit. Overlap was shared aligned residues divided by candidate-interval length. A candidate was excluded when overlap exceeded 0.5 and the canonical domain score was at least the candidate score minus 5 bits. The highest-scoring retained candidate was used when multiple candidates remained.

Family construction

Retained proteins were clustered with MMseqs2 at minimum identity 0.30 and coverage 0.60, coverage mode 0, cluster mode 1. Clusters required occurrence in at least five genomes, median length below 250 amino acids, at least 60 percent of members below 250 amino acids, and absence of clearly nonregulatory annotation keywords.

Nonregulatory exclusion list

A cluster was excluded when its dominant Pfam label exactly matched or began with one of the following patterns: eIF-5_eIF-2B, Ribosomal_S19e, TFIIE_alpha, SRP19, TFIIE_beta, RNA_pol_Rpb, S10_plectin, Ribosomal_L29, Ribosomal_S13, tRNA_bind, RuvA_N, RuvA_C, DNA_pol, Rad51, RecA, DEAD, Helicase_C, ResIII, HNH, GIY-YIG, Homeodomain, Pou, Ets, Fork_head, CBFD_NFYB_HMF, Myb_DNA-binding, Myb_DNA-bind, HTH_psq, zf-, Integrase, rve, Recombinase, Resolvase, Terminase, Phage, DNA_binding_1, Sigma54_activat, Sigma70_r2, Sigma70_r3, Sigma70_r1, AAA, Cdc6_C, Mor, CsrA, KH, OB_, RRM, dsrm, PAZ or Piwi. This rule removes obvious non-target families but cannot establish regulatory activity for retained clusters.

Catalog sequence recovery

Protein sequences for all 13,237 released family members were recovered by accession. Versioned NCBI assembly protein FASTA files supplied 13,232 sequences; five accessions absent from their current assembly files were recovered through NCBI Protein. Every recovered sequence matched the length recorded in the membership table. The release includes a deterministic compressed FASTA, source URLs, retrieval dates, source-file checksums, accession-level fallback records and an empty missing-sequence report.

Panel and projection sensitivity

Five hundred seeded within-panel resamples tested three alternative sampling schemes using fixed family membership. A lineage-stratified accumulation analysis sampled 10 to 100% of each major group in 10% increments, with 500 replicates below the complete panel. A separate 500-replicate analysis sampled 331 lineage-matched genomes from the independent 3,192-genome projection matrix after removing discovery-panel accessions. These analyses test fixed-catalog observation, retention and HMM transfer, not de novo rediscovery.

Genome-quality sensitivity

The 3,192 accession-resolved projection genomes were stratified by origin, assembly level, and reported completeness/contamination. Per-family prevalence and detected-family counts were reported. Contrasts are descriptive because strata are confounded with lineage and sampling.

Taxonomic clustered prevalence

Organism names defined 533 inferred genus clusters. Genera were resampled with replacement for 1,000 seeded replicates to obtain genus-balanced and cluster-weighted prevalence intervals. This addresses cluster sampling but is not a species-tree model.

Motif holdout

For twelve families, PWMs were fitted on taxonomically separated training data. Genome-blocked splits held out genomes, genus-blocked splits held out complete genera, and the strictest analysis left out one major lineage at a time with same-lineage other-family controls. Minimum sample-size failures were reported as unevaluable. Results remain predictive rather than biochemical.

Catalog-wide complete-genus family separability

All 13,237 deposited proteins were represented by normalized dipeptide frequencies. Each test sequence was classified against family centroids fitted after every sequence from its inferred genus had been removed. Families required at least two remaining training sequences. The analysis evaluated 13,166 sequences from 245 families and reports sequence assignments, family-level exact binomial intervals, separation margins, dominant family-pair errors and macro- and micro-accuracy. Assignment to the most abundant eligible training family and uniform assignment among eligible families provide descriptive baselines. This is a complete-genus composition check, not calibrated HMM validation.

Evidence limits

Neighborhood enrichment is treated as association, motif prediction as sequence-level prediction, and structure matching as similarity to a curated reference set. None alone demonstrates direct DNA binding or transcriptional regulation.

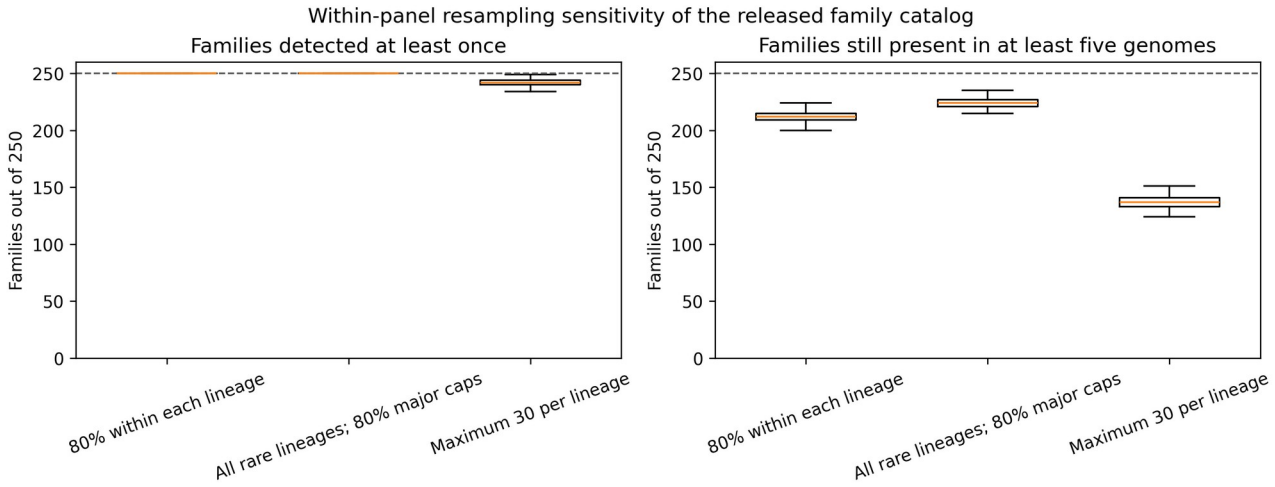
Supplementary Table S1 Analysis classification

Analysis	Role	Question	Principal limit
Catalog size and membership	Primary descriptive	Prespecified released pipeline summaries	Bounded to two Pfam clans
Held-out family HMM assignment	Secondary validation	Withheld sequence assignment	Random split; possible taxonomic leakage
Independent-genome projection	Secondary validation	Zero accession overlap	Fixed HMMs; no rediscovery
Genome/genus/lineage-blocked motif AUC	Robustness	Out-of-sample discrimination	Predictive, not biochemical
Genome-quality stratification	Sensitivity	Robustness across deposited strata	Descriptive; confounded
Genus-clustered prevalence	Sensitivity	Taxonomic sampling imbalance	Organism-name genera; no species tree
Catalog-wide complete-genus separability	Secondary validation	Taxonomically blocked boundary coherence	Composition-based; five families unevaluable
Neighborhood enrichment	Exploratory	Functional hypothesis generation	No direct target claim

Supplementary Table S2 Data artifact definitions

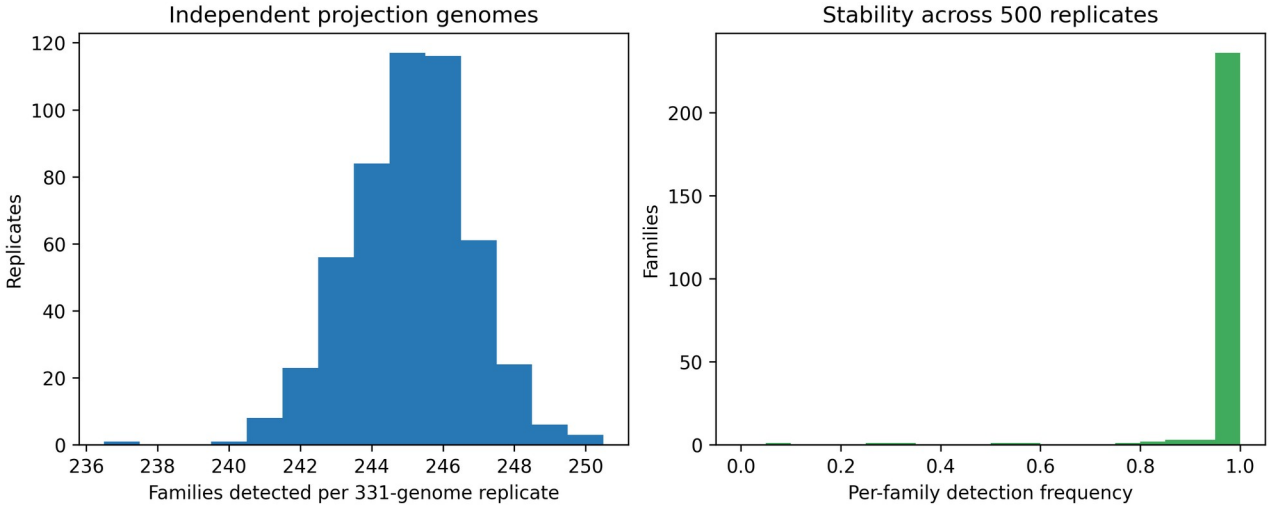
Artifact	Scope	Permitted interpretation
Discovery panel	331 accessions	Family discovery and fixed-member sensitivity
Legacy projection matrix	250 × 3,192	Accession-resolved projection sensitivity
Final genome accounting	3,526 genomes	Archived aggregate fates and denominators
Corrected prevalence	250 families × 7 groups	Archived group-level values; final family matrices unavailable
Family upstreams	12 families; 250-nt sequences	Genome/genus/lineage-blocked motif validation
Catalog member sequences	13,237 proteins; 250 families	Complete-genus composition validation

Supplementary Figure S1



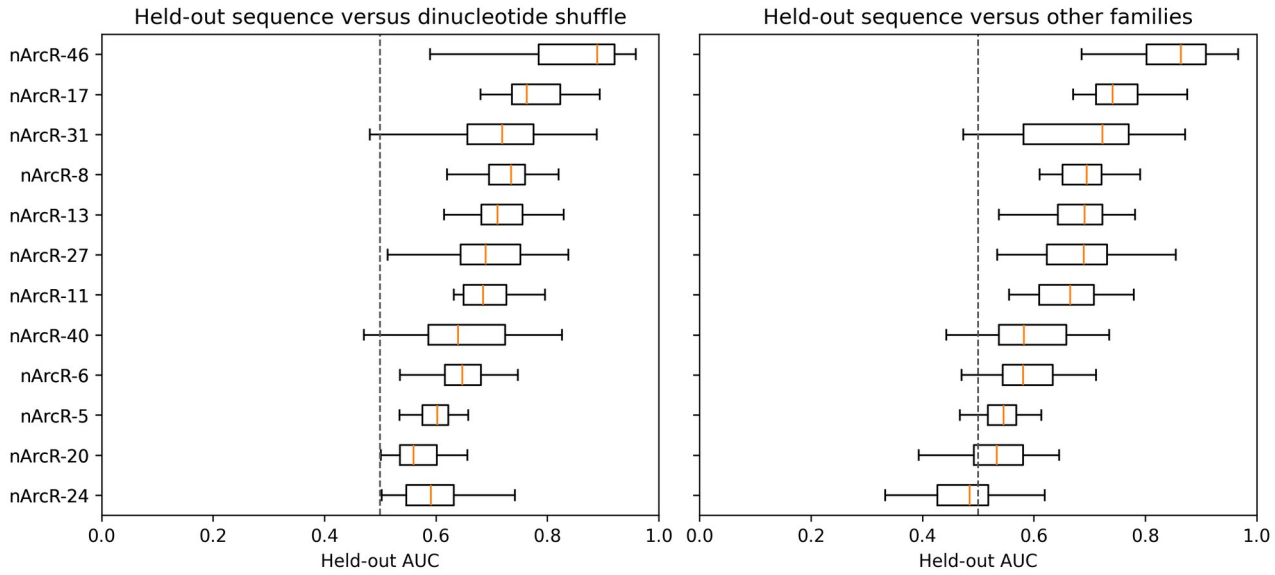
Supplementary Figure S1. Within-panel fixed-family retention under three alternative sampling schemes. Boxes summarize 500 seeded replicates. This analysis asks whether already-defined families remain represented; it is not a de novo clustering sensitivity analysis.

Supplementary Figure S2



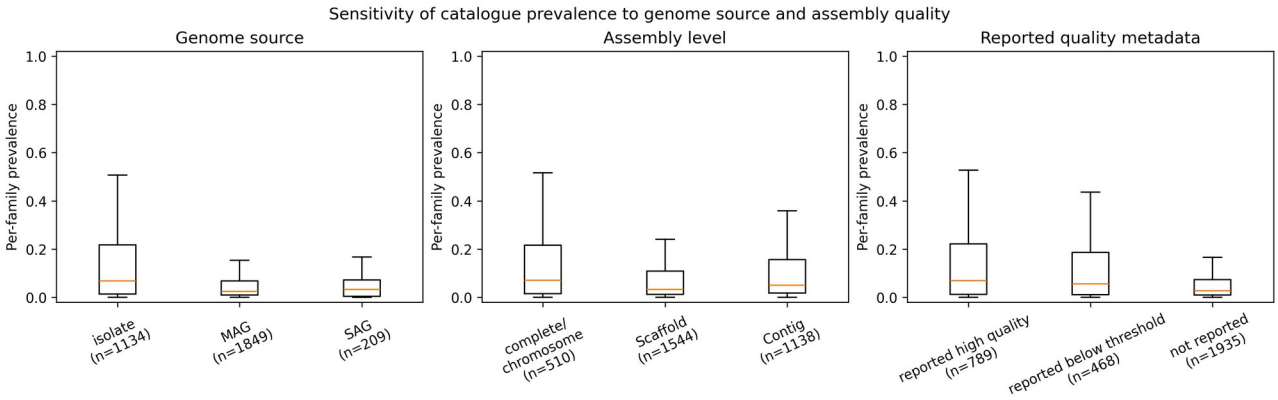
Supplementary Figure S2. Projection of released family HMMs to 500 lineage-matched samples of 331 independent genome accessions. The discovery/projection accession overlap is zero. Results test HMM transfer, not independent family discovery.

Supplementary Figure S3



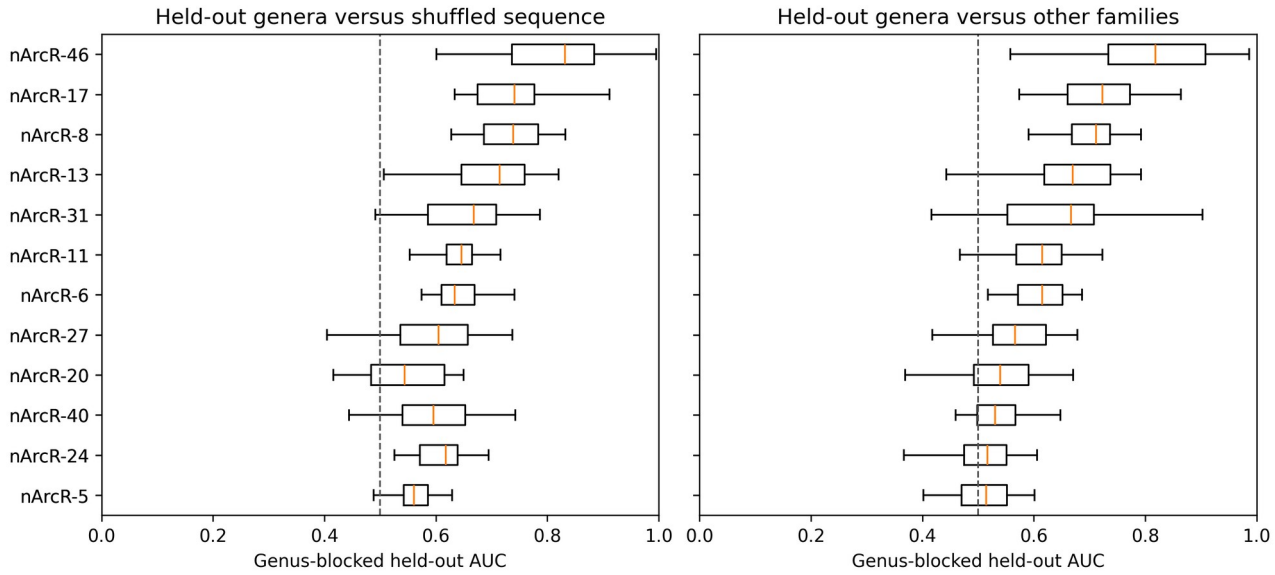
Supplementary Figure S3. Genome-level motif holdout for 12 families. PWMs were fitted only to training genomes and scored against held-out family upstreams, dinucleotide shuffles, and GC-matched other-family upstreams. Repeated splits overlap and are interpreted descriptively.

Supplementary Figure S4



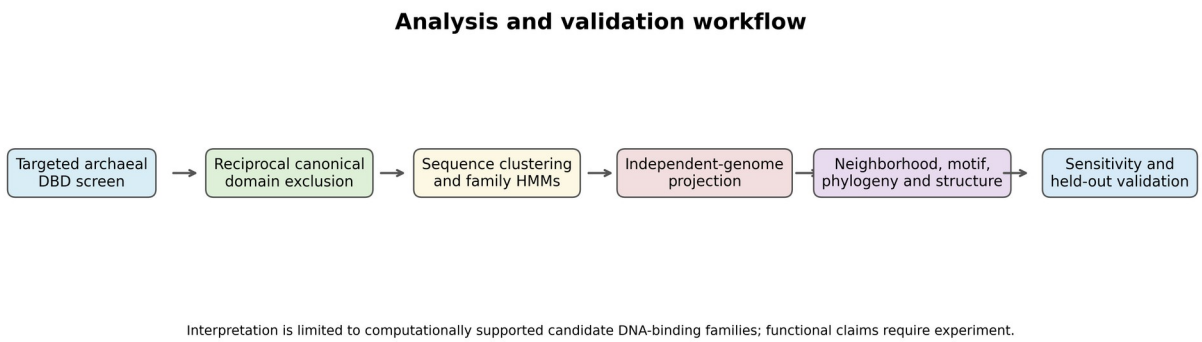
Supplementary Figure S4. Sensitivity of per-family prevalence to genome source, assembly level, and reported completeness/contamination. Boxes summarize the 250 family prevalence values within each stratum. Differences are descriptive because strata are confounded with lineage and sampling.

Supplementary Figure S5



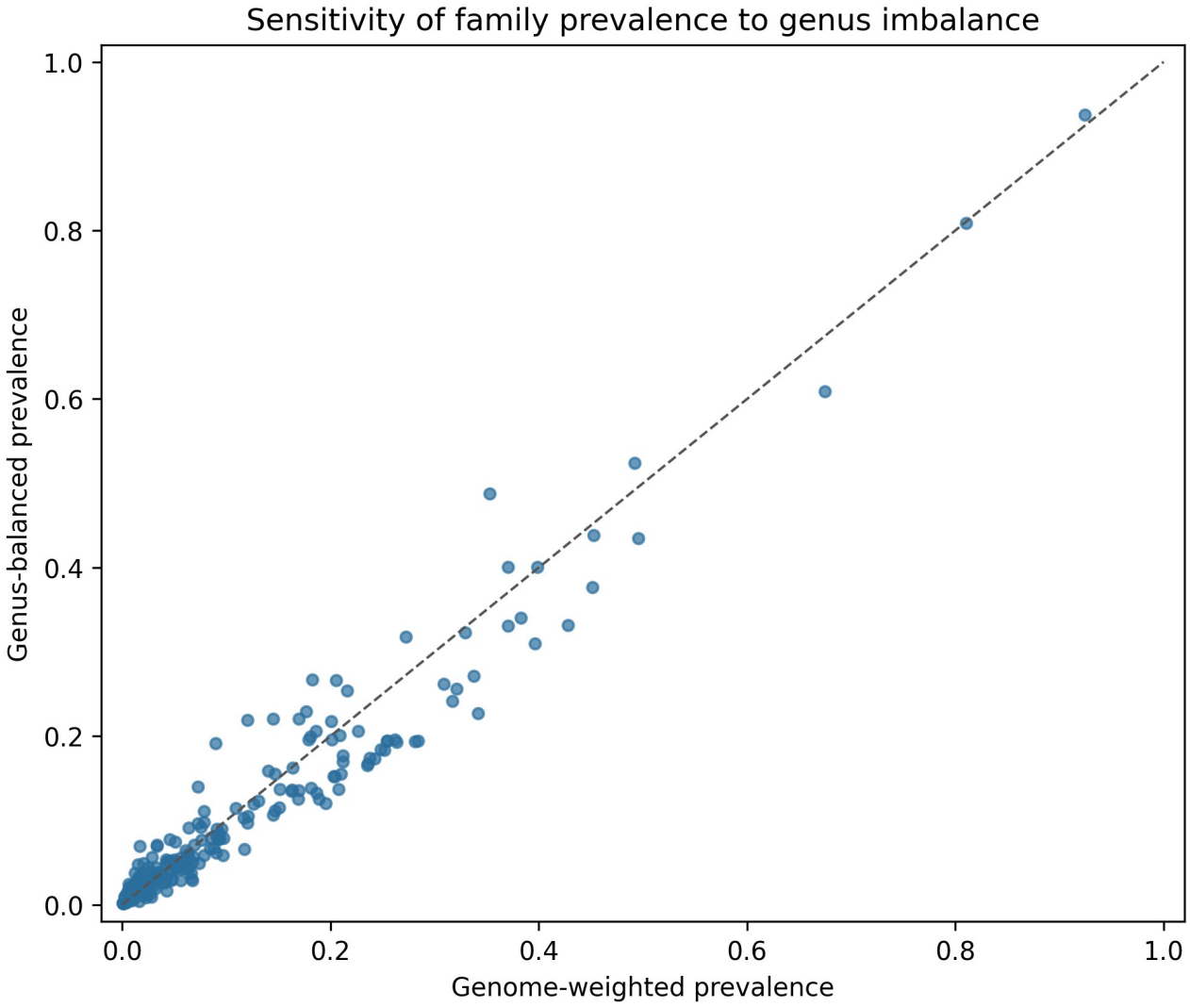
Supplementary Figure S5. Genus-blocked motif holdout with zero train/test genus overlap. Seven of 12 families exceeded AUC 0.5 against other-family controls in at least 26 of 30 resamples. This stricter result shows that some genome-level signal is lineage-associated.

Supplementary Figure S6



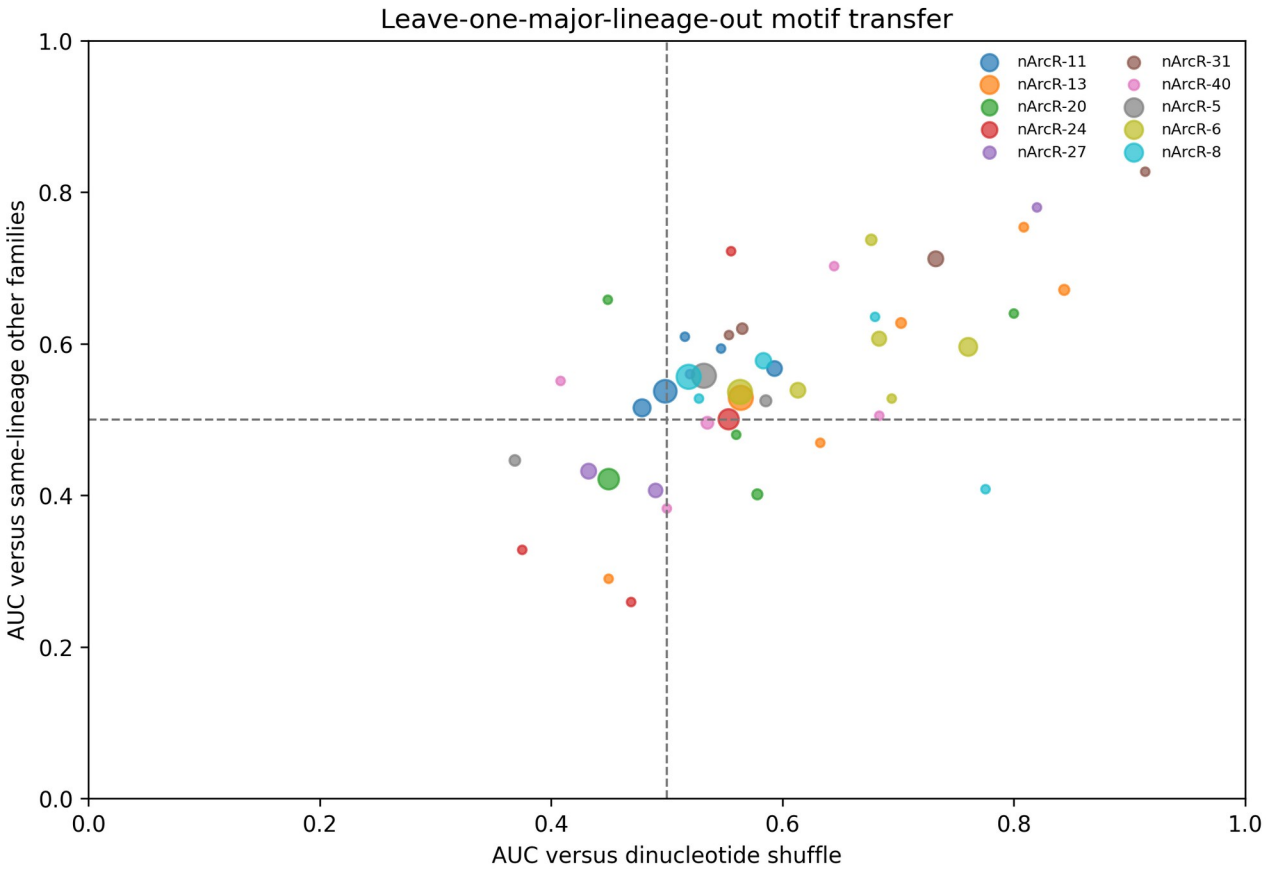
Supplementary Figure S6. Workflow from the targeted two-clan scan through family construction, projection, contextual analyses, and sensitivity validation. The endpoint is a bounded candidate-family catalog, not an experimentally confirmed regulome.

Supplementary Figure S7



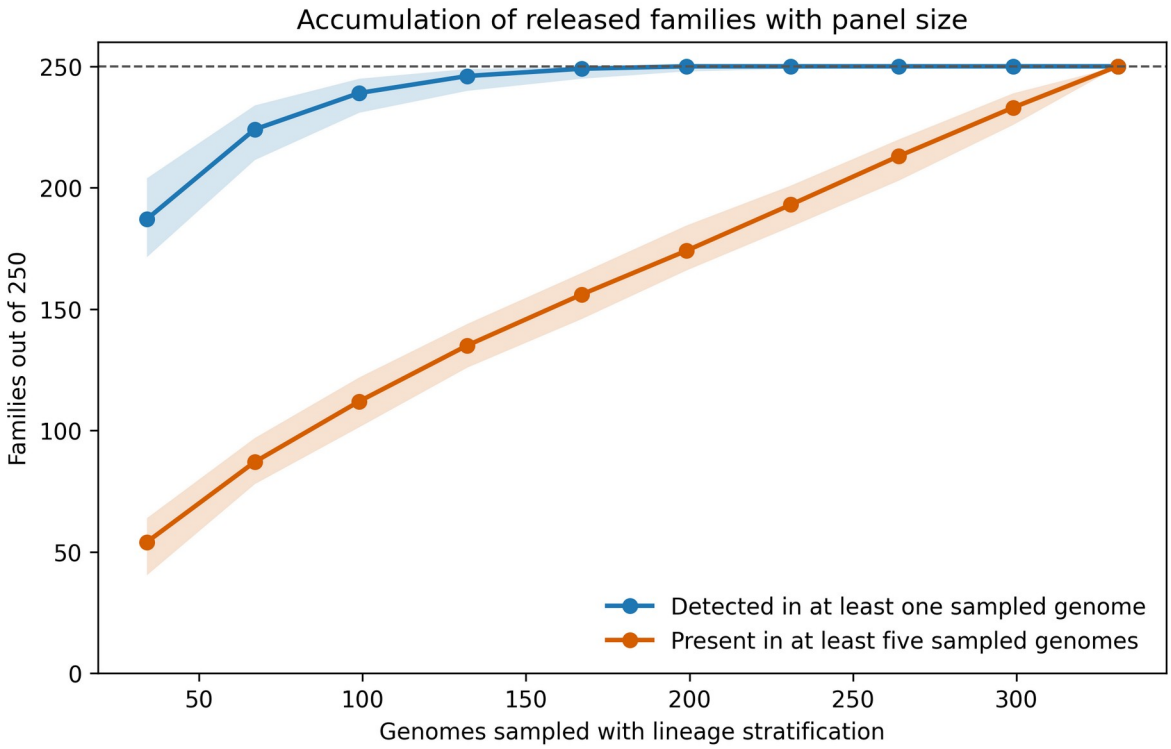
Supplementary Figure S7. Comparison of genome-weighted and genus-balanced prevalence for 250 families. Genus clusters were inferred from deposited organism names. Deviations from the diagonal show sensitivity to uneven genus representation.

Supplementary Figure S8



Supplementary Figure S8. Leave-one-major-lineage-out motif transfer. Point size reflects held-out sequence count. Only blocks satisfying prespecified target, training, and same-lineage control minima are plotted; two families have no evaluable block.

Supplementary Figure S9



Supplementary Figure S9. Lineage-stratified accumulation of the released families as panel size increases. Lines show medians and shaded regions show 95% resampling intervals. The blue curve counts families observed at least once; the orange curve counts families represented in at least five sampled genomes. This is a fixed-catalog sensitivity analysis and does not repeat candidate scanning or de novo clustering.

Supplementary Table S3 Search universe provenance

Component	Reported count	Identifiers available	Status
Target Pfam clans	2	CL0123; CL0057	partially documented
Reference families	19	no	unrecoverable historical metadata
Reference profile models	20	no	unrecoverable historical metadata
Candidate profile models	747	no	unrecoverable historical metadata
Released catalog family HMMs	250	yes	deposited

The derived 250-family HMM library is deposited, but it does not replace the unavailable historical Pfam profile inventory. No profile accession or database release has been inferred retrospectively.

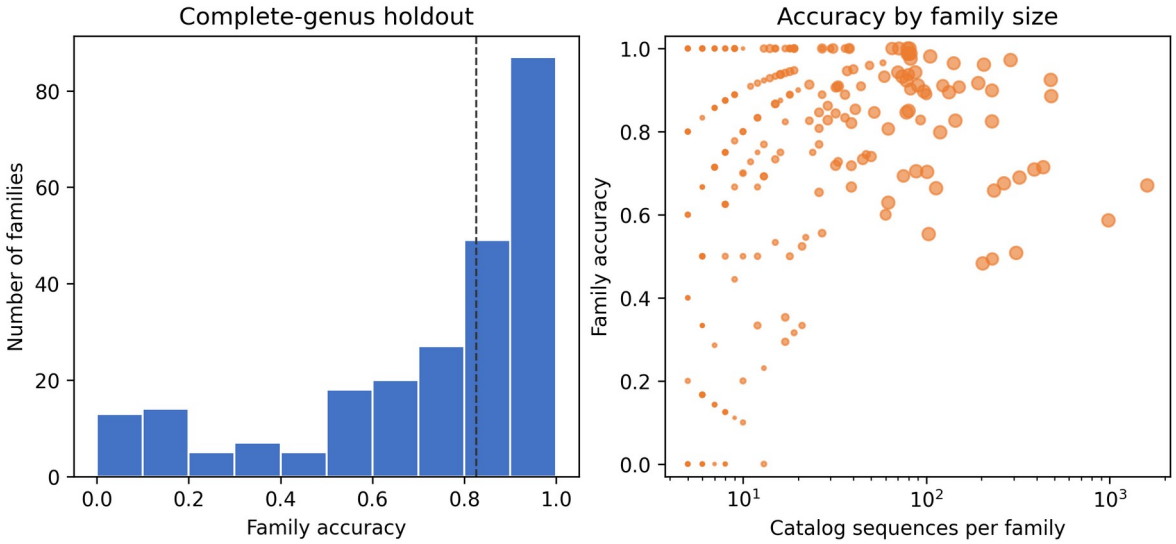
Supplementary Table S4 Taxonomic group mapping

Display label	Discovery genomes	Accounting genomes	Definition limit
Euryarchaeota	140	2378	Historical major-group assignment; exact GTDB release unavailable
Crenarchaeota within TACK	70	270	Mutually exclusive historical major-group assignment
Thaumarchaeota within TACK	51	283	Mutually exclusive historical major-group assignment
Other deep TACK	15	302	Excludes Crenarchaeota, Thaumarchaeota and Korarchaeota categories
Korarchaeota within TACK	3	7	Mutually exclusive historical major-group assignment
DPANN	33	228	Historical major-group assignment; exact GTDB release unavailable
Asgardarchaeota	19	58	Historical major-group assignment; exact GTDB release unavailable

Supplementary Table S5 Catalog-wide complete-genus validation

Measure	Value
Catalog families	250
Evaluable families	245
Evaluable sequences	13166
Micro-accuracy	0.767
Macro family accuracy	0.721
Median family accuracy	0.826
Descriptive accuracy baselines	majority training family 0.122; uniform chance 0.004
Families with accuracy below 0.5	44
Largest directional confusion	nArcR-2 to nArcR-12, 246 sequences

Supplementary Figure S10



Supplementary Figure S10. Catalog-wide complete-genus holdout using normalized dipeptide-composition centroids. The left panel shows family-level accuracy for 245 evaluable families; the dashed line marks the median of 0.826. The right panel relates accuracy to catalog size, with point area proportional to evaluable genera. Micro-accuracy was 0.767, compared with 0.122 for the majority-training-family baseline and 0.004 for uniform assignment. Forty-four families had accuracy below 0.5. This analysis reduces genus-level leakage but does not calibrate HMM thresholds or repeat family discovery.